





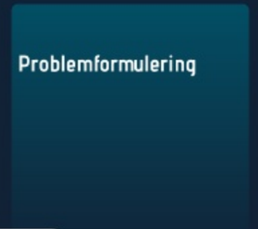


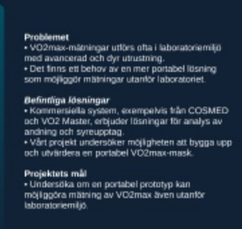


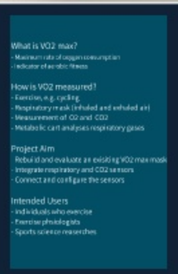


Rebuilding a VO2 max Mask Prototype

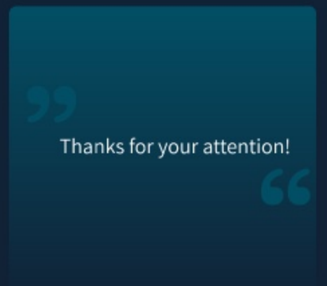
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What is VO2 max?

- Maximum rate of oxygen consumption
- Indicator of aerobic fitness

How is VO2 measured?

- Exercise, e.g. cycling
- Respiratory mask (inhaled and exhaled air)
- Measurement of O_2 and CO_2
- Metabolic cart analyses respiratory gases

Project Aim

- Rebuild and evaluate an exisiting VO2 max mask
- Integrate respiratory and CO_2 sensors
- Connect and configure the sensors

Intended Users

- Individuals who exercise
- Exercise phsiologists
- Sports science reaserches



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Problemformulering

Problemet
• VO2max-mätningar utförs ofta i laboratoriemiljö med avancerad och dyr utrustning.
• Det finns ett behov av en mer portabel lösning som möjliggör mätningar utanför laboratoriet.

Befintliga lösningar
• Kommersiella system, exempelvis från COSMED och VO2 Master, erbjuder lösningar för analys av andning och syreupptag.
• Vårt projekt undersöker möjligheten att bygga upp och utvärdera en portabel VO2max-mask.

Projektets mål
• Undersöka om en portabel prototyp kan möjliggöra mätning av VO2max även utanför laboratoriemiljö.

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Målet är att bygga en bärbar prototyp





Detta görs genom att miniatyrisera den redan existerande tekniken!

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Prototypens teknik

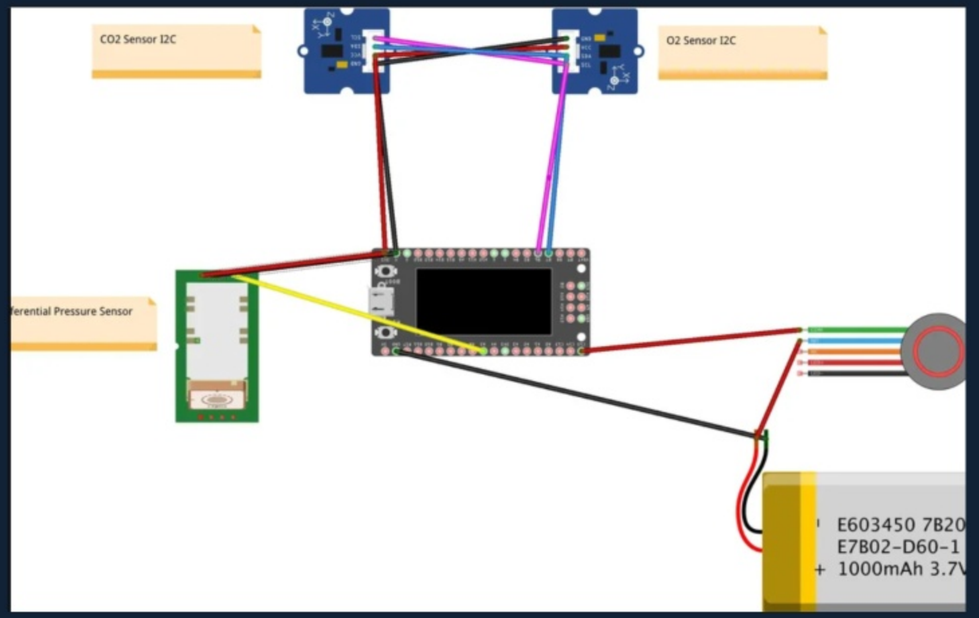
De grundlägagnde aspekterna:

Rör som användaren andas genom

Sensorer som analyserar innehållet

Mikrokontroller som beräknar fram VO2 MAX

Batteri som försörjer systemet.



De ingående komponenterna:

Venturi-rör

Elektrokemisk O2-sensor

Termisk CO2-sensor

Differential trycksyensor

ESP32 mikrokontroller

3.7 V LiPo batteri

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User Interaction

Two roles. One workflow.
From setup to data review.



01

Prepare and calibrate

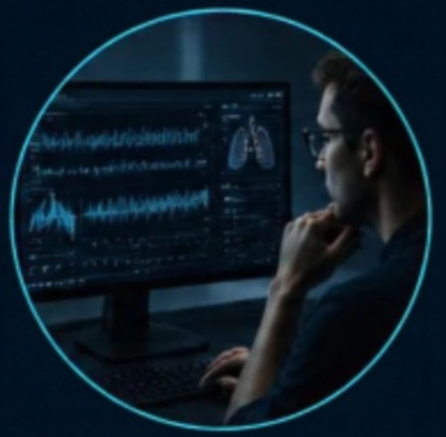
Researcher



02

Collect breathing data

Participant



03

View readings and examine changes

Researcher

"Illustration of how the system is intended to be used."

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Results

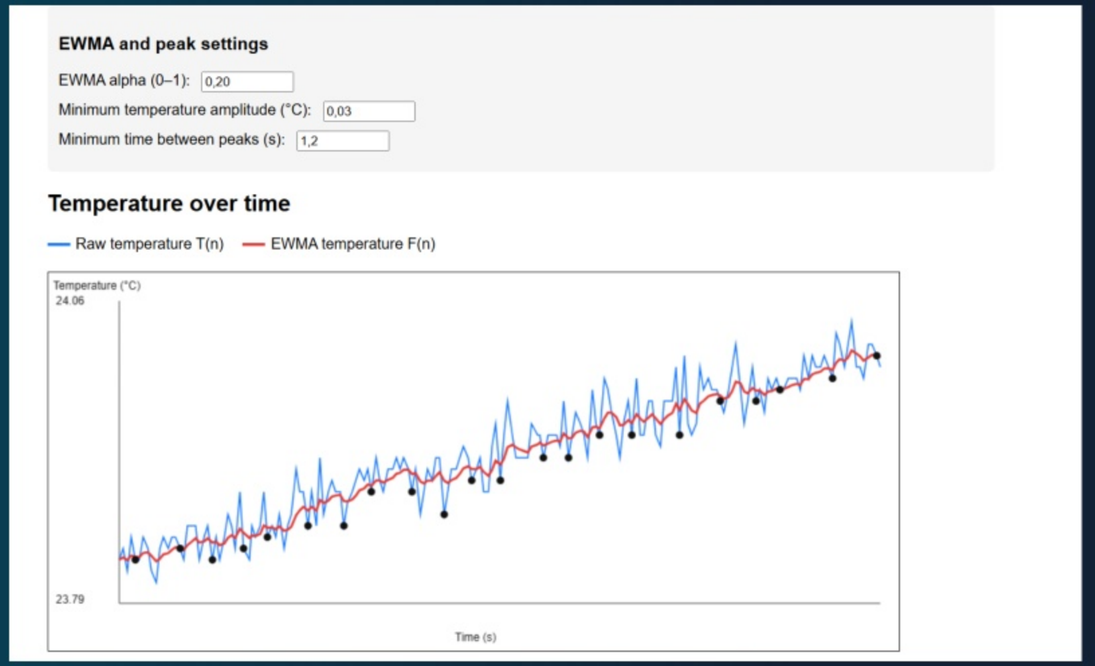
clear progress, remaining gaps, and what blocks validation

Förbättrade sensor ansutningar och utvecklade en webbplats

Använde EWMA-filter för att identifiera andningscykler och frekvens

Koldioxid värderna var opålitliga och kommunikationsfel kvarstod

3d-printade delar färdigställdes, men kalibrering och valdering återstår




```
14:49:01.491 -> CO2 reading failed
14:49:01.706 -> CO2 reading failed
14:49:01.889 -> CO2 reading failed
14:49:02.101 -> CO2 reading failed
14:49:02.314 -> CO2 reading failed
14:49:02.479 -> CO2 reading failed
14:49:02.693 -> CO2 reading failed
14:49:02.899 -> CO2 reading failed
14:49:03.122 -> CO2 reading failed
14:49:03.297 -> CO2 reading failed
14:49:03.506 -> CO2 reading failed
```



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Project Methodology

01 reviewed inherited design

02 repaired connections

03 debugged and tested

04 continued troubleshooting

WHAT WE'RE DOING DIFFERENTLY NEXT TIME

Secure components and calibration tools **earlier**

Test sensors individually **before** integration

Record working **software versions**



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Thanks for your attention!

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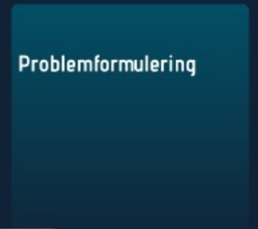


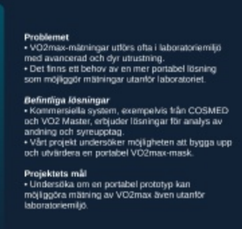


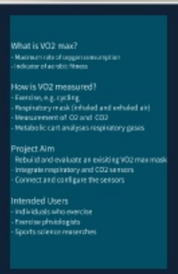


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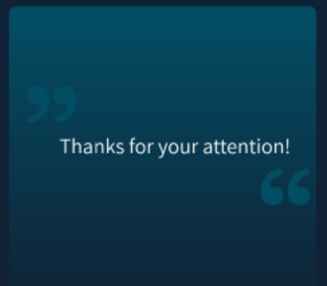
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